

Chapter 2. Second-order Linear Differential Equations

Sections 2.1, 2.2, 2.3, 2.4, 2.5

1. Consider the second-order linear homogeneous differential equation

$$\frac{d^2y}{dt^2} + p(t)\frac{dy}{dt} + q(t)y = 0. \quad (1)$$

Let $y_1(t)$ and $y_2(t)$ be two solutions of (1). The wronskian of $y_1(t)$ and $y_2(t)$ is defined as:

$$W(t) = W[y_1, y_2](t) = y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

- $W[y_1, y_2](t)$ is either identically zero, or is never zero.
 - $y_1(t)$ and $y_2(t)$ are linearly independent on the interval $\alpha < t < \beta$ if and only if $W[y_1, y_2](t) \neq 0$, i.e., $y_1(t)$ and $y_2(t)$ form a fundamental set of solutions of (1).
2. Consider the second-order linear homogeneous differential equation with constant coefficients:

$$L[y] = a\frac{d^2y}{dt^2} + b\frac{dy}{dt} + cy = 0. \quad (2)$$

We need to find two linearly independent solutions $y_1(t)$ and $y_2(t)$ of (2), so that the general solution of (2) is of the form

$$y(t) = c_1y_1(t) + c_2y_2(t).$$

Characteristic equation:

$$ar^2 + br + c = 0.$$

Find its two roots r_1 and r_2 .

- If r_1 and r_2 are real and $r_1 \neq r_2$, then

$$y_1(t) = e^{r_1t}, \quad y_2(t) = e^{r_2t}.$$

- If r_1 and r_2 are complex (automatically $r_1 = \bar{r}_2$), i.e., $r_1 = \alpha + i\beta$ and $r_2 = \alpha - i\beta$, then

$$y_1(t) = e^{\alpha t} \cos(\beta t), \quad y_2(t) = e^{\alpha t} \sin(\beta t).$$

- If $r_1 = r_2$ (automatically real), then

$$y_1(t) = e^{r_1t}, \quad y_2(t) = te^{r_1t}.$$

3. Consider the initial-value problem:

$$a\frac{d^2y}{dt^2} + b\frac{dy}{dt} + cy = 0; \quad y(t_0) = y_0, \quad y'(t_0) = y_0'. \quad (3)$$

To solve it, first find a general solution $y(t) = c_1y_1(t) + c_2y_2(t)$ and then determine the two constants c_1 and c_2 using the initial conditions.

4. Consider the second-order linear equation

$$L[y] = \frac{d^2y}{dt^2} + p(t)\frac{dy}{dt} + q(t)y = 0. \quad (4)$$

If $y_1(t)$ is known to be a solution of (4), then another linearly independent solution $y_2(t)$ can be found by the **method of reduction of order**:

$$y_2(t) = y_1(t)v(t) = y_1(t) \int u(t) dt$$

where

$$u(t) = \frac{e^{-\int p(t) dt}}{y_1^2(t)}$$

5. Consider the second-order linear nonhomogeneous differential equation:

$$L[y] = \frac{d^2y}{dt^2} + p(t)\frac{dy}{dt} + q(t)y = g(t). \quad (5)$$

The general solution of (5) must be of the form

$$y(t) = c_1y_1(t) + c_2y_2(t) + \psi(t)$$

where $y_1(t)$ and $y_2(t)$ are two linearly independent solutions of the homogeneous equation $L[y] = 0$ and $\psi(t)$ is a particular solution of (5).

- Using the **method of variation of parameters**, we can get

$$\psi(t) = u_1(t)y_1(t) + u_2(t)y_2(t)$$

where

$$u_1(t) = \int -\frac{g(t)y_2(t)}{W[y_1, y_2](t)} dt,$$

$$u_2(t) = \int \frac{g(t)y_1(t)}{W[y_1, y_2](t)} dt.$$

6. Consider the second-order linear nonhomogeneous differential equation with constant coefficients:

$$L[y] = a\frac{d^2y}{dt^2} + b\frac{dy}{dt} + cy = g(t). \quad (6)$$

To find a particular solution $\psi(t)$ of (6) using judicious guessing.

- Case I: $g(t) = a_0 + a_1t + \cdots + a_nt^n$

$$\psi(t) = \begin{cases} A_0 + A_1t + \cdots + A_nt^n, & c \neq 0 \\ t(A_0 + A_1t + \cdots + A_nt^n), & c = 0, b \neq 0 \\ \frac{1}{a} \left[\frac{a_0t^2}{1 \cdot 2} + \frac{a_1t^3}{2 \cdot 3} + \cdots + \frac{a_nt^{n+2}}{(n+1)(n+2)} \right], & c = b = 0 \end{cases}$$

- Case II: $g(t) = e^{\alpha t}(a_0 + a_1t + \cdots + a_nt^n)$
 Characteristic equation: $ar^2 + br + c = 0$
 (i) α is not a root of the characteristic equation, then

$$\psi(t) = (A_0 + A_1t + \cdots + A_nt^n)e^{\alpha t}$$

- (ii) α is a single root of the characteristic equation, then

$$\psi(t) = t(A_0 + A_1t + \cdots + A_nt^n)e^{\alpha t}$$

- (iii) α is a double root of the characteristic equation, then

$$\psi(t) = e^{\alpha t}v(t)$$

and $L[\psi(t)] = e^{\alpha t}av''(t)$, and $v''(t) = \frac{1}{a}g(t)$.

- Case II: $g(t) = (a_0 + a_1t + \cdots + a_nt^n) \times \begin{cases} \cos(\omega t) \\ \sin(\omega t) \end{cases}$
 Let $\phi(t) = u(t) + iv(t)$ be a particular solution of the equation of

$$L[y] = a\frac{d^2y}{dt^2} + b\frac{dy}{dt} + cy = (a_0 + a_1t + \cdots + a_nt^n)e^{i\omega t}.$$

Then

$$\psi(t) = \begin{cases} u(t) = \operatorname{Re}\{\phi(t)\} \\ v(t) = \operatorname{Im}\{\phi(t)\} \end{cases}$$

repectively.

7. Consider the second-order linear nonhomogeneous differential equation with constant coefficients:

$$L[y] = a\frac{d^2y}{dt^2} + b\frac{dy}{dt} + cy = \sum_{i=1}^k g_i(t). \quad (7)$$

Then a particular solution $\psi(t)$ of (7) is given by

$$\psi(t) = \sum_{i=1}^k \psi_i(t)$$

where $\psi_i(t)$ is a particular solution of $L[y] = g_i(t)$.